Staying Power: The Individual Effects of the H-1B Visa Lottery

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Motivation

- Large literature: effects of immigration on U.S. labor market [Card 1990; Borjas 2003; Ottaviano and Peri 2012]

- Common setting: **H-1B visa program**
  - Tied to employer-worker pair; allocated *via lottery*
  - Recent concerns about fraud + rising demand $\rightarrow$ new \$100k fee

- Existing literature: GE, local labor market, firm effects [Bound et al. 2017; Doran et al. 2022; Mahajan et al. 2024]
  - How are firms using the lottery in practice?
  - What happens to workers who lose the lottery?

- **This Project:** Estimating **worker-level effects** of winning H-1B lottery

Linkage Strategy

Empirical hurdles in estimating worker-level H-1B effects:

1. Lack of information on H-1B lottery losers or activity outside U.S.
   - **This Project:** New data on universe of H-1B applicants + LinkedIn profiles
2. No personal identifiers (name, ID number) in H-1B data
   - **This Project:** Probabilistic linking to LinkedIn profile data across many sparse fields (including firm)
3. Don't observe workers' outcomes unless they win visa
   - Traditional view of H-1B: firm hires worker if they win visa, otherwise don't
   - In reality, majority of H-1B applicants are *already working at firm* on a different temporary visa
     - New view of H-1B: firm employs worker who stays if they win visa, otherwise leaves
   - **This Project:** Link by subsetting to workers already at firm at time of application

Result: Individual-level panel data on 65k H-1B lottery applicants

This Paper

Research Question: What are the effects of winning the H-1B lottery on the individual worker?

1. Estimate effects of lottery win on indiv. outcomes
   - **Preview:** Winning ↑ U.S. retention by 1.16 pp (1.2%); firm retention by 2.64 pp (3.6%) 1 yr post-lottery
   - Suggestive evidence of job stagnation for winners (stay at same firm but less likely to get promotion)
   - Losers do not reallocate to new education
2. Investigating robustness and heterogeneity: why are effects small?
   1. **Measurement Error:** Estimated effects 3-4x larger in higher-confidence linked samples, but still small
   2. **Student visa:** Effects largest for workers near student work authorization expiration, but still small

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Empirical Puzzle: How are lottery losers managing to stay?

Literature Review

- **Effects of H-1B Visa**
  - Aggregate and firm-level impacts on native employment and wages [Kerr et al. 2015; Peri et al. 2015a; Doran et al. 2022; Mahajan et al. 2024], innovation and productivity [Kerr and Lincoln 2010; Peri et al. 2015b; Dimmock et al. 2022; Mayda et al. 2023], and employer monopsony power [Kim and Pei 2022]
  - Global effects on offshoring [Glennon 2024], origin countries [Khanna and Morales 2023], and third-party countries [Brinatti and Guo 2024]
  - **This paper:** first individual-level evidence
- **International Students and U.S. Labor Markets**
  -
  - Offshoring, Indian IT industry, immigration to third countries [Glennon 2024; Khanna and Morales 2023; Brinatti and Guo 2024]
  - Brain drain and return migration [Amanzadeh et al. 2024; Adda et al. 2022]
  - **This paper:** quantifies effect of H-1B cap on individual workers globally
- **Probabilistic Record Linkage** [Fellegi and Sunter 1969; Bailey et al. 2020; Enamorado

Outline

1. Introduction
2. Background and Context
3. Data and Linkage
4. Results
5. Conclusion

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What is an H-1B Visa?

- Temporary work visa (6 years) for workers in 'specialty occupations'
  - Requires highly specialized knowledge + at least a bachelor's degree
- Employer-sponsored: firm must apply on worker's behalf
  - Portable between employers, but no self-sponsorship
  - Employer can sponsor worker for green card (but application tied to employer)
- Capped at 85,000 visas per year since 2004
  - 65,000 regular cap + 20,000 advanced degree exempt (ADE) for U.S. M.A.+
  - Demand has exceeded cap every year since 2013
- Since 2013: visas allocated via **lottery** each spring
  - Results in April, visa valid starting October

The H-1B Lottery: Key Facts

- FY2021: USCIS moved to electronic registration ($10 fee vs. $500+ full petition)
  - **Critical for this paper:** USCIS began keeping records of lottery *losers*
  - Prior data: only winners observed
- Electronic system enabled fraud: duplicate applications via different employers
  - Outsourcing/staffing firms 'overapply' — treat workers as interchangeable
  - I exclude firm-years with any duplicate applicants
- Win rates: ∼35–40% for regular lottery
  - ADE applicants entered into separate lottery first (higher win rate)
- **Identification:** within firm-year, lottery selection is as good as random

Optional Practical Training: The Key Bridge

- F-1 (student) visa holders may work after graduation under OPT
  - Standard OPT: 12 months of authorized employment
  - STEM extension: up to 36 months total for STEM graduates
- OPT has become dominant pathway into H-1B
  - 50–70% of recent H-1B approvals go to workers already in U.S.
  - 94% of F-1→H-1B transitions use OPT
- **Common trajectory:** graduate → OPT → apply for H-1B each spring (up to 3 tries)
  - Worker already employed at sponsoring firm during OPT
  - Firm retains worker if they win; worker faces uncertainty if they lose
- **Key insight:** pre-lottery employment relationship enables linkage

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Data Sources

- **H-1B Application Data** (FOIA via Bloomberg, 2020–2023)
  - All applicants: country of birth, birth year, sex, employer name/location
  - Winners only: current visa type, education level/field, job title/salary
  - 1.8 million applications; first dataset with both winners and losers
- **Revelio Labs LinkedIn Data**
  - Profiles (894m individuals): full name, education history, work history
- **Challenge:** H-1B data is deidentified — no names, no unique identifiers
  - 'Needle in a haystack' problem without employer
- **Key insight:** restrict candidates to workers already at sponsoring firm before lottery

Overview of Linkage Strategy

1. **Employer crosswalk:** fuzzy match H-1B firms to Revelio firms (93% match rate)
   - Exclude firm-years with 10+ applications or any duplicate applicants
   - Iterative process: deterministic matching → LLM review
2. **Revelio imputation:** estimate country of birth, birth year, sex
   - For now: simple ML model using Wikipedia data (name2nat) + deterministic rules
   - Future: LLM-based review using full profile information (description, etc.)
3. **Candidate scoring:** for each applicant, pick 'close-enough' candidates matches *working at sponsoring firm* at time of lottery
   - Restrict to LinkedIn users with US education (OPT eligible) but no US high school (not native)
   - Score candidates on match quality

Result: weighted panel of 167,598 linked applications

Sample Variants

- **Baseline:** keep unrestricted sample, collapse at application level (N = 70,580)
- **One-to-one** (preferred): optimal bipartite matching (N = 64,781)
  - Each applicant assigned to at most one Revelio candidate (and vice versa) within a lottery year
- **Strict (Q25 / Q50 / Q75):** score threshold samples
  - Restrict to applicants whose best match score exceeds the 25th, 50th, or 75th percentile
  - Higher confidence matches; trade off sample size for precision
- **Rare Country:** exclude high-multiplicity countries (N = 10,440)
  - Drop applicants from Southern Asia, East Asia, UK, Canada, Australia
  - Lower candidate pool multiplicity → sharper individual identification

Can evaluate match quality using H-1B winners: for now overly strict Match Quality Table

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Empirical Specification

For applicant i at firm c in lottery year t, estimate:

$$y_{ict}^{s} = \beta_0 + \beta_1 Win_{ict} + \beta_2 ADE_{it} + \gamma_{ct} + \varepsilon_{ict}^{s}$$

- Win_{ict}: lottery selection indicator
- ADE_{it}: advanced degree exempt (separate, higher-odds lottery)
- γ_{ct}: firm-by-lottery-year fixed effects
- Outcomes measured s years post-lottery; weights from probabilistic linkage
- Standard errors clustered by firm
- Alternate specification: include Win $\times$ ADE interaction (results similar)

Identification: within firm-year, lottery selection random conditional on ADE

Outcome Variables

- **Location:** In U.S., in home country, in non-U.S. non-home country
- **Employment:** still at same firm, new position conditional on same firm (proxy for promotion)
- **Education:** new education (possible channel for staying in U.S.)

Winning the Lottery ↑ U.S. Presence...

	In U.S.	In home ctry	Non-US non-home	Same firm	New pos. (cond.)	New educ.
	(1)	(2)	(3)	(4)	(5)	(6)
winner	0.0116^{***}	-0.0060^{***}	-0.0033^{***}			
	(0.0027)	(0.0015)	(0.0011)			
Control mean	0.950	0.019	0.010			
N	64,781	64,781	64,781			

OLS estimates; one-to-one linked sample (N = 64,781); firm-by-lottery-year FE; SEs clustered by firm. $t = 1$ year post-lottery. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Additional outcomes

...and ↑ Job Stability (or stagnation)?

	In U.S.	In home ctry	Non-US non-home	Same firm	New pos. (cond.)	New educ.
	(1)	(2)	(3)	(4)	(5)	(6)
winner	0.0116^{***}	-0.0060^{***}	-0.0033^{***}	0.0264^{***}	-0.0146^{*}	
	(0.0027)	(0.0015)	(0.0011)	(0.0053)	(0.0075)	
Control mean	0.950	0.019	0.010	0.743	0.448	
N	64,781	64,781	64,781	64,781	45,209	

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Losers do not reallocate to new education

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	(0.0027)	(0.0015)	(0.0011)	(0.0053)	(0.0075)	(0.0059)
Control mean	0.950	0.019	0.010	0.743	0.448	0.340
N	64,781	64,781	64,781	64,781	45,209	64,781

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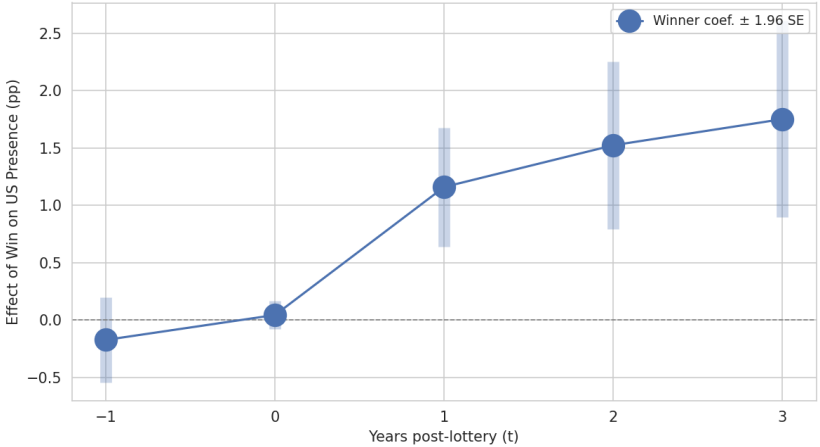
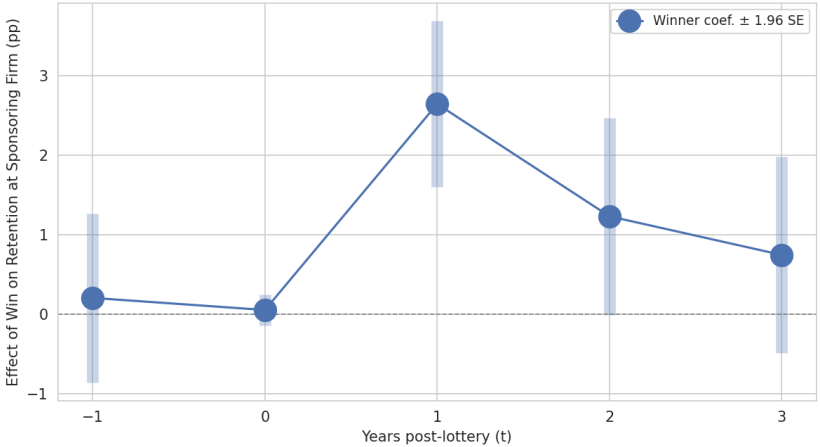
Effect on U.S. Presence Grows Over Time



Raw means

But Firm Retention Effect Peaks in Year 1



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Taking Stock: Why Are Effects Small?

- Effects are statistically significant but moderate in the one-to-one sample (1.16 pp ↑ U.S. presence)
- **Reason 1:** Measurement error attenuates estimates
  - Probabilistic linkage mixes correct and incorrect matches
  - **How do effects vary by linkage quality?**
- **Reason 2:** OPT workers have fallback options
- **Reason 3:** Firms have other ways to retain workers
- **Reason 4:** Differential profile updating

Effects Robust Across Higher-Confidence Linked Samples

	In U.S.	Same firm
	(1)	(2)
One-to-one ($N = 64{,}781$)	0.0116^{***} (0.0027)	0.0264^{***} (0.0053)

$\hat{\beta}_1$ *(winner effect) with SEs in parentheses; firm-by-lottery-year FE; SEs clustered by firm.* $t = 1$ *year post-lottery.*
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Control means

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Baseline ($N = 70{,}580$)	0.0023^{***}	(0.0008)	0.0058^{***}	(0.0017)
Strict Q25 ($N = 40{,}165$)	0.0103^{***}	(0.0026)	0.0253^{***}	(0.0048)
Strict Q50 ($N = 27{,}765$)	0.0146^{***}	(0.0034)	0.0328^{***}	(0.0060)
Strict Q75 ($N = 13{,}894$)	0.0142^{***}	(0.0049)	0.0408^{***}	(0.0086)

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Rare Country ($N = 10{,}440$)	0.0083^{**}	(0.0033)	0.0221^{***}	(0.0061)

$\hat{\beta}_1$ (winner effect) with SEs in parentheses; firm-by-lottery-year FE; SEs clustered by firm. $t = 1$ year post-lottery.
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Control means

Taking Stock: Why Are Effects Small?

- Effects are statistically significant but moderate in the one-to-one sample (1.16 pp U.S. presence)

- **Reason 1:** Measurement error attenuates estimates
  - **Verdict:** Partial driver, but estimates still small in high-confidence samples

- **Reason 2:** OPT workers...use OPT?
  - Selecting on OPT $\implies$ applicants can continue to work through OPT!
  - **Prediction:** effects increase as OPT runs out
  - **How do effects vary by years since graduation?**

- **Reason 3:** Firms have other ways to retain workers

- **Reason 4:** Differential profile updating

Effects Strongest for Workers Near OPT Expiration

	Dependent Variable: In U.S.			
	0 yrs post-grad	1 yr post-grad	2 yrs post-grad	3 yrs post-grad
	(1)	(2)	(3)	(4)
winner	0.0089	0.0150	0.0158	0.0520***
	(0.0129)	(0.0094)	(0.0099)	(0.0174)
Control mean	0.946	0.927	0.952	0.918
N	1,980	6,029	4,728	3,043

Outcome: In U.S. (1 year post-lottery). Cols (1)–(4): subgroups by years since graduation at time of lottery application.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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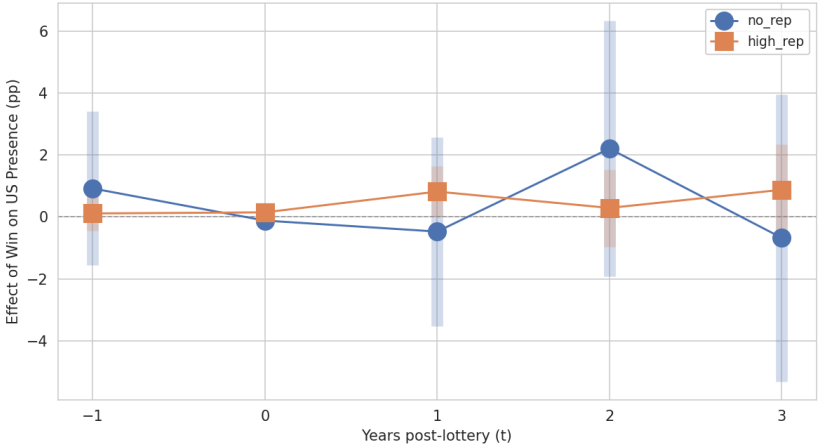
- **Reason 2:** OPT workers...use OPT?
  - **Verdict:** Effects largest as applicants near OPT expiration, but still small even for those 3 years post-grad
- **Reason 3:** Firms have other ways to retain workers
  - Firms can re-sponsor lottery losers in subsequent years
  - **Prediction:** effects smaller at firms more likely to re-sponsor (*high-rep*)
  - **How do effects vary by employer type?**
- **Reason 4:** Differential profile updating

Sample Sizes Too Small to Detect Effects



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  - **Verdict:** Effects largest as applicants near OPT expiration, but still small even for those 3 years post-grad
- **Reason 3:** Firms have other ways to retain workers
  - **Verdict:** Sample sizes too small
- **Reason 4:** Differential profile updating
  - Winners may update LinkedIn more actively, inflating observed U.S. presence
  - **Prediction:** winners show more activity updates; location effects driven by profile changes, not actual stays
  - **How does profile updating vary by lottery outcome?**

Winners Update Profiles Less not More

	Activity update diff (1)	Profile in U.S. (2)	Profile home country (3)	Profile non-U.S. (4)	Profile loc. null (5)
winner	1.1155**				
	(0.4624)				
Control mean	17.01				
N	64,712				

OLS estimates; one-to-one linked sample; firm-by-lottery-year FE; SEs clustered by firm. $t = 1$ year post-lottery. Cols (2)–(5): profile location outcomes. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

And Location Effects Hold With Alternate Measure

	Activity update diff	Profile in U.S.	Profile home country	Profile non-U.S.	Profile loc. null
	(1)	(2)	(3)	(4)	(5)
winner	1.1155^{**}	0.0152^{***}	-0.0087^{***}	-0.0143^{***}	-0.0009
	(0.4624)	(0.0038)	(0.0024)	(0.0032)	(0.0022)
Control mean	17.01	0.885	0.051	0.083	0.031
N	64,712	64,781	64,781	64,781	64,781

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Conclusion

- **Main effects:** winning H-1B raises U.S. presence (+1.16 pp) and firm retention (+2.64 pp)
- **Magnitude of effects:** not fully explained by measurement error or OPT
- **Policy Implication/Puzzle:** OPT workers managing to remain without H-1B — how?

Next Steps:

- Explore more mechanisms: underreporting on LinkedIn, L-1 (intra-firm transfer), heterogeneity in school result
- Firm-level effects: benchmarking my estimates with literature

Thank You!

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Additional Outcomes: One-to-One Sample

	Profile in U.S.	Cont. education	Activity update diff	Frac. null end date
	(1)	(2)	(3)	(4)
winner	—	0.0040	0.7620^{***}	-0.0129^{**}
	—	(0.0059)	(0.2922)	(0.0056)
Control mean	—	0.663	17.011	0.408
N	—	64,781	64,781	64,781

OLS estimates; one-to-one linked sample (N = 64,781); firm-by-lottery-year FE; SEs clustered by firm. $t = 1$ year post-lottery. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Return

Robustness: Control Means by Sample

	In U.S.	Same firm
One-to-one	0.950	0.743
Baseline	0.956	0.743
Strict Q25	0.938	0.741
Strict Q50	0.934	0.740
Strict Q75	0.937	0.746
Rare Country	0.948	0.749

Control group means; $t = 1$ year post-lottery.

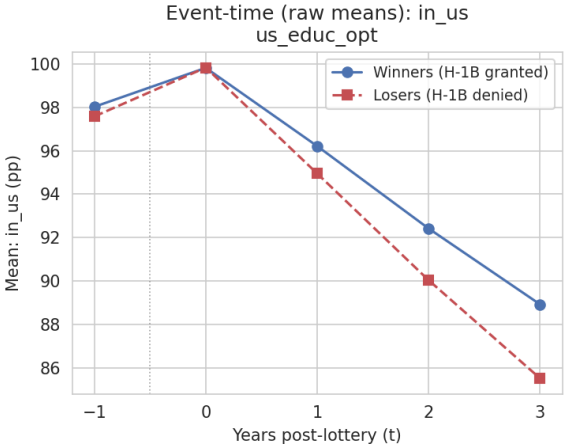
Return

Higher-Quality Samples Have Better Match Validation

Dimension	FOIA cov.	U.S.-educ.	Pre-filtered	One-to-one	Strict Q50
Education level	26–33%	62.0%	61.7%	73.9%	85.7%
Institution (ADE)	—	34.4%	45.9%	46.6%	49.2%
Field exact	100%	19.6%	20.0%	27.9%	33.7%
Job title	100%	4.0%	3.7%	6.1%	9.0%
Ed. + any other	—	34.9%	34.6%	47.1%	60.3%
N (applications)		106,819	25,231	101,797	47,881

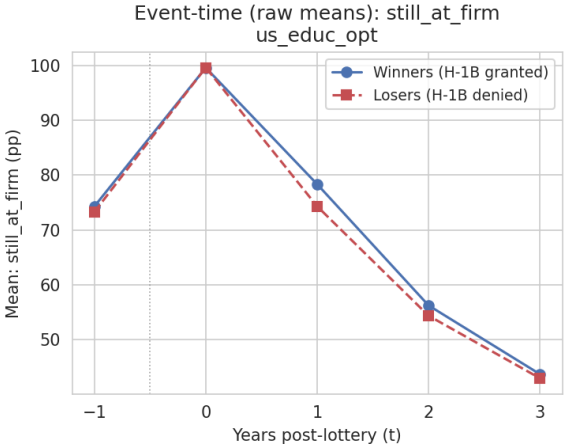
Agreement rates between FOIA petition fields and matched Revelio profiles, by sample.

Raw Means: U.S. Presence by Lottery Outcome



Return

Raw Means: Firm Retention by Lottery Outcome



Return